

Complete Answer & Marking Guide

All four tutorials — Permutations & Combinations, Relations & Graphs, Primes & GCD, and Number Theory — with exhaustive step-by-step solutions, diagrams, and explanations.

TUTORIAL 1

Permutations, Combinations & Binomial Coefficients

QUESTION 1

How many permutations are there of the set $\{a, b, c, d, e, f, g\}$?

► MODEL ANSWER

- 1 A permutation of a set is an ordered arrangement of ALL its elements. For a set of n elements, the number of permutations is $n!$ (n factorial).
- 2 The set has $n = 7$ elements.
- 3 Calculate:

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$$

♦ $7! = 5040$ permutations

QUESTION 2

Let $S = \{1, 2, 3, 4, 5\}$. (a) List all *three-element* permutations. (b) List all *three-element* combinations.

► MODEL ANSWER

- 1 **Key distinction:** A *permutation* is ordered ($ABC \neq BAC$). A *combination* is unordered ($ABC = BAC = CAB$).
- 2 Number of 3-permutations: $P(5, 3) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{120}{2} = 60$. Number of 3-combinations: $C(5, 3) = \binom{5}{3} = \frac{5!}{3!2!} = 10$.
- 3 (a) The 10 three-element combinations (unordered — listing combinations first is easier, then permutations derive from them):

Combination	All permutations (6 each)
{1, 2, 3}	123, 132, 213, 231, 312, 321
{1, 2, 4}	124, 142, 214, 241, 412, 421
{1, 2, 5}	125, 152, 215, 251, 512, 521
{1, 3, 4}	134, 143, 314, 341, 413, 431
{1, 3, 5}	135, 153, 315, 351, 513, 531
{1, 4, 5}	145, 154, 415, 451, 514, 541
{2, 3, 4}	234, 243, 324, 342, 423, 432
{2, 3, 5}	235, 253, 325, 352, 523, 532
{2, 4, 5}	245, 254, 425, 452, 524, 542
{3, 4, 5}	345, 354, 435, 453, 534, 543

♦ (a) 60 three-element permutations (all rows in the "permutations" column above) | (b) 10 three-element combinations (left column)

QUESTION 3

How many possibilities for win, place, show (1st, 2nd, 3rd) in a 12-horse race?

► MODEL ANSWER

- 1 We are selecting 3 horses from 12 and assigning them to ordered positions (1st, 2nd, 3rd). Order matters → **Permutation**.

- 2

$$P(12, 3) = \frac{12!}{(12-3)!} = \frac{12!}{9!} = 12 \times 11 \times 10$$

3

$$12 \times 11 = 132, \quad 132 \times 10 = 1320$$

♦ $P(12, 3) = 1320$ possible outcomes

QUESTION 4

A group has n men and n women. How many ways to arrange them in a row so that men and women alternate?

► MODEL ANSWER

- ① There are $2n$ people total. For alternating gender, the pattern must be M W M W ... or W M W M ...
- ② **Pattern 1: M W M W M W ...** (starts with man)
 - Arrange n men in the odd positions: $n!$ ways
 - Arrange n women in the even positions: $n!$ ways
 - Subtotal: $n! \times n!$
- ③ **Pattern 2: W M W M W M ...** (starts with woman)
 - By the same logic: $n! \times n!$
- ④ Total = $n! \cdot n! + n! \cdot n! = 2 \cdot (n!)^2$

♦ $2 \cdot (n!)^2$ arrangements

QUESTION 5

How many subsets with an *odd* number of elements does a 10-element set have?

► MODEL ANSWER

- ① A 10-element set has $2^{10} = 1024$ subsets in total.
- ② Key identity: for any set with n elements, exactly half the subsets have odd size and half have even size. This follows from the binomial identity $\sum_{k \text{ odd}} \binom{n}{k} = 2^{n-1}$.
- ③ We can verify: odd-sized subsets of a 10-element set are those of size 1, 3, 5, 7, 9:

$$\binom{10}{1} + \binom{10}{3} + \binom{10}{5} + \binom{10}{7} + \binom{10}{9} = 10 + 120 + 252 + 120 + 10 = 512$$

④ $512 = 2^{10-1} = 2^9 \checkmark$

♦ $2^9 = 512$ subsets with an odd number of elements

QUESTION 6

A coin is flipped 10 times. How many outcomes: (a) total, (b) exactly 2 heads, (c) at most 3 tails, (d) same number of heads and tails?

► MODEL ANSWER

① **Total outcomes:** Each flip has 2 outcomes. By the product rule: $2^{10} = 1024$

② **Exactly 2 heads:** Choose 2 of the 10 flips to be heads:

$$\binom{10}{2} = \frac{10!}{2!8!} = \frac{10 \times 9}{2} = 45$$

③ **At most 3 tails:** means 0, 1, 2, or 3 tails (equivalently, at least 7 heads):

$$\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} = 1 + 10 + 45 + 120 = 176$$

④ **Same number of heads and tails:** Need exactly 5 heads and 5 tails:

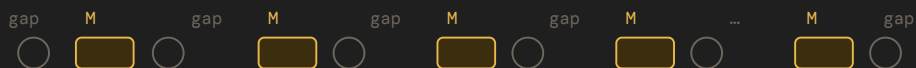
$$\binom{10}{5} = \frac{10!}{5!5!} = 252$$

♦ (a) 1024 | (b) 45 | (c) 176 | (d) 252

QUESTION 7

How many ways for 8 men and 5 women to stand in a line so that no two women stand next to each other?

- ① **Step 1 — Place the 8 men:** Arrange the 8 men in a row. There are $8!$ ways.
- ② **Step 2 — Identify gaps:** With 8 men placed, there are 9 available slots (gaps) where women can stand without being adjacent to each other — one before each man, one after each man:



8 men create 9 available gaps (including ends). 5 women must each go in a different gap.

- ③ **Step 3 — Place the 5 women in the gaps:** Choose 5 of the 9 gaps and arrange the 5 women in those gaps. Order matters (women are distinct), so this is $P(9, 5)$:

$$P(9, 5) = \frac{9!}{(9-5)!} = \frac{9!}{4!} = 9 \times 8 \times 7 \times 6 \times 5 = 15120$$

- ④ **Total:**

$$8! \times P(9, 5) = 40320 \times 15120 = 609,638,400$$

♦ $8! \times P(9, 5) = 40320 \times 15120 = 609,638,400$

QUESTION 8

13 people show up (3 are women). (a) Ways to choose 10 players? (b) Ways to assign 10 positions from 13? (c) Ways to choose 10 with at least 1 woman?

- a) **Choose 10 from 13 (order does NOT matter for just fielding a team):**

$$\binom{13}{10} = \binom{13}{3} = \frac{13 \times 12 \times 11}{3!} = \frac{1716}{6} = 286$$

- b) **Assign 10 positions (order matters — each position is distinct):**

$$P(13, 10) = \frac{13!}{(13-10)!} = \frac{13!}{3!} = 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4$$

$$= 1,037,836,800$$

(c) At least 1 woman — use complementary counting:

$$\text{At least 1 woman} = \text{Total} - \text{No women}$$

$$\text{No women} = \binom{10}{10} = 1 \quad (\text{only 10 men, choose all 10})$$

$$\text{Answer} = 286 - 1 = 285$$

♦ (a) 286 | (b) 1,037,836,800 | (c) 285

QUESTION 9

40 T/F questions, 17 are true. How many different answer keys are possible?

► MODEL ANSWER

(1) An answer key is determined by choosing which 17 of the 40 positions get answer "True". The rest are "False". Order of the *questions* matters for telling them apart, but we are choosing a *subset* of positions.

(2)

$$\binom{40}{17} = \frac{40!}{17! 23!}$$

(3) Computing: $\binom{40}{17} = 18,643,560$

♦ $\binom{40}{17} = 18,643,560$ **different answer keys**

QUESTION 10

7 women and 9 men. Committee of 5. (a) At least 1 woman. (b) At least 1 woman AND 1 man.

► MODEL ANSWER

(1) Total ways to choose 5 from 16: $\binom{16}{5} = \frac{16 \times 15 \times 14 \times 13 \times 12}{120} = 4368$

a At least 1 woman = Total – No women:

$$\text{No women (all men)} = \binom{9}{5} = 126$$

$$\text{At least 1 woman} = 4368 - 126 = 4242$$

b At least 1 woman AND 1 man = Total – (All women) – (All men):

$$\text{All women} = \binom{7}{5} = 21$$

$$\text{All men} = \binom{9}{5} = 126$$

$$\text{Answer} = 4368 - 21 - 126 = 4221$$

♦ (a) 4242 | (b) 4221

QUESTION 11

Strings of 6 lowercase English letters. How many contain: (a) letter a, (b) letters a and b, (c) "ab" consecutively with all distinct, (d) a somewhere left of b, all distinct?

► MODEL ANSWER

Alphabet has 26 letters. Repetition is allowed unless stated otherwise.

a Contains letter a (repetition allowed): Use complement: Total strings – strings with NO 'a':

$$26^6 - 25^6 = 308,915,776 - 244,140,625 = 64,775,151$$

b Contains both a and b (repetition allowed): By inclusion-exclusion:

$$\begin{aligned} & 26^6 - (\text{no a}) - (\text{no b}) + (\text{no a and no b}) \\ &= 26^6 - 25^6 - 25^6 + 24^6 \\ &= 308,915,776 - 244,140,625 - 244,140,625 + 191,102,976 \\ &= 11,737,502 \end{aligned}$$

c "ab" consecutive (as a block), all letters distinct: Treat "ab" as one block → we have 5 entities to arrange (the block + 4 other distinct letters). – Choose 4 more distinct letters from the 24

remaining (not a or b): $P(24, 4)$ ways to pick and arrange them. – Place the "ab" block in one of the 5 positions among the 5 entities: 5 positions. Wait — simpler: treat "ab" as a single unit. We have this unit plus 4 other letters from remaining 24 (distinct). The 5 slots for a 5-unit string:

$$\underbrace{5}_{\text{positions for "ab" block}} \times P(24, 4) = 5 \times 24 \times 23 \times 22 \times 21 = 5 \times 255,024 = 1,275,120$$

- (d) a somewhere to the left of b, all letters distinct:** First count strings of 6 distinct letters containing both a and b:

$$P(24, 4) \times \binom{6}{2} \times 1 \quad \text{— choose 4 others, then place a and b in 2 of 6 slots with a left of b}$$

More directly: Take all 6-letter strings with distinct letters containing both a and b:

$$\binom{24}{4} \times 4! \times 6! / (6 - 6)!$$

Simplest: choose 4 other distinct letters from 24: $\binom{24}{4}$ ways. Arrange all 6 letters: $6!$ arrangements. Of these, exactly half have a to the left of b (by symmetry).

$$\frac{\binom{24}{4} \times 6!}{2} = \frac{10626 \times 720}{2} = \frac{7,650,720}{2} = 3,825,360$$

♦ (a) 64,775,151 | (b) 11,737,502 | (c) 1,275,120 | (d) 3,825,360

QUESTION 12

Department: 10 men, 15 women. Committee of 6 with more women than men.

► MODEL ANSWER

- (1)** "More women than men" in a committee of 6 means: 4W+2M, 5W+1M, or 6W+0M.

(2) 4 women, 2 men: $\binom{15}{4} \binom{10}{2} = 1365 \times 45 = 61,425$

5 women, 1 man: $\binom{15}{5} \binom{10}{1} = 3003 \times 10 = 30,030$

6 women, 0 men: $\binom{15}{6} \binom{10}{0} = 5005 \times 1 = 5,005$

(3) Total = $61,425 + 30,030 + 5,005 = 96,460$

♦ 96,460 committees with more women than men

Q13: Expand $(x + y)^5$. Q14: Coefficient of x^5y^8 in $(x + y)^{13}$. Q15: Coefficient of $x^{101}y^{99}$ in $(2x - 3y)^{200}$.

► MODEL ANSWER

Binomial Theorem: $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$

13 Expand $(x + y)^5$:

$$\begin{aligned} (x + y)^5 &= \sum_{k=0}^5 \binom{5}{k} x^{5-k} y^k \\ &= \binom{5}{0} x^5 + \binom{5}{1} x^4 y + \binom{5}{2} x^3 y^2 + \binom{5}{3} x^2 y^3 + \binom{5}{4} x y^4 + \binom{5}{5} y^5 \\ &= x^5 + 5x^4 y + 10x^3 y^2 + 10x^2 y^3 + 5x y^4 + y^5 \end{aligned}$$

14 Coefficient of x^5y^8 in $(x + y)^{13}$: The general term is $\binom{13}{k} x^{13-k} y^k$. We need $13 - k = 5$ so $k = 8$.

$$\text{Coefficient} = \binom{13}{8} = \binom{13}{5} = \frac{13 \times 12 \times 11 \times 10 \times 9}{120} = 1287$$

15 Coefficient of $x^{101}y^{99}$ in $(2x - 3y)^{200}$: Write as $(2x + (-3y))^{200}$. General term:

$$\binom{200}{k} (2x)^{200-k} (-3y)^k = \binom{200}{k} 2^{200-k} (-3)^k x^{200-k} y^k$$

Need $200 - k = 101$ so $k = 99$.

$$\text{Coefficient} = \binom{200}{99} \cdot 2^{101} \cdot (-3)^{99}$$

Since $(-3)^{99} = -3^{99}$ (odd power):

$$\text{Coefficient} = -\binom{200}{99} \cdot 2^{101} \cdot 3^{99}$$

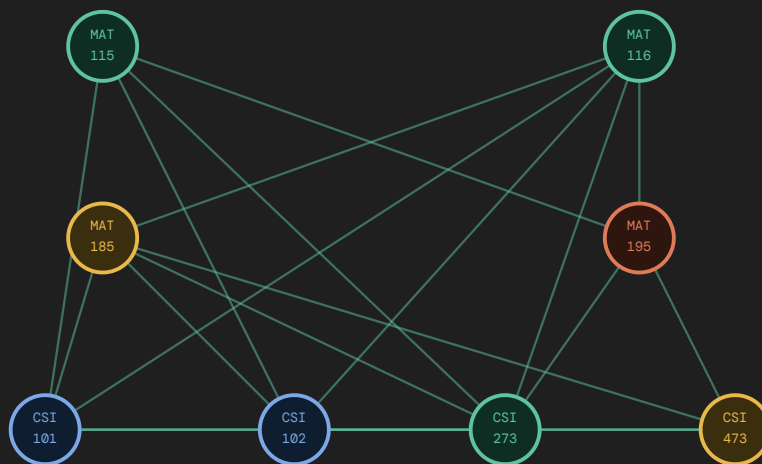
♦ Q13: $x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$ | Q14: 1287 | Q15: $-\binom{200}{99} \cdot 2^{101} \cdot 3^{99}$

QUESTION 1

Exam scheduling conflict graph for 8 courses. Draw conflict graph, colour it minimally, give adjacency list.

► MODEL ANSWER

- 1 **Conflict = shared students (edge between courses).** The NON-conflicts listed are the missing edges. All other pairs are connected.
- 2 **Non-edges (pairs with NO shared students):**
 - MAT115 – CSI473
 - MAT116 – CSI473
 - MAT195 – CSI101
 - MAT195 – CSI102
 - MAT115 – MAT116
 - MAT115 – MAT185
 - MAT185 – MAT195
- 3 **Conflict graph (edges exist for all other pairs):**



Conflict graph. Node colour = exam time slot (see colouring below). Edges = shared students.

- 4 **Proper colouring (chromatic number analysis):** Look for cliques (fully connected subsets). MAT115, MAT195, CSI101 are mutually connected — form a triangle (clique of 3) → chromatic number ≥ 3 . We

can achieve 4 colours:

Colour / Time Slot	Courses
Colour 1 (Teal)	MAT115, CSI273
Colour 2 (Amber)	MAT116, CSI473
Colour 3 (Terracotta)	MAT185, CSI101
Colour 4 (Blue)	MAT195, CSI102

Verify: no two courses of the same colour share an edge. ✓

5 Adjacency list:

Course	Adjacent to (conflicts with)
MAT115	MAT195, CSI101, CSI102, CSI273
MAT116	MAT185, MAT195, CSI101, CSI102, CSI273
MAT185	MAT116, CSI101, CSI102, CSI273, CSI473
MAT195	MAT115, MAT116, CSI273, CSI473
CSI101	MAT115, MAT116, MAT185, CSI102, CSI273, CSI473
CSI102	MAT115, MAT116, MAT185, CSI101, CSI273, CSI473
CSI273	MAT115, MAT116, MAT185, MAT195, CSI101, CSI102, CSI473
CSI473	MAT185, MAT195, CSI101, CSI102, CSI273

✦ Chromatic number = 4. Four exam time slots are sufficient (and necessary).

QUESTION 2

6 radio stations. Conflict if within 150 miles. Find conflict graph, minimal colouring, adjacency list.

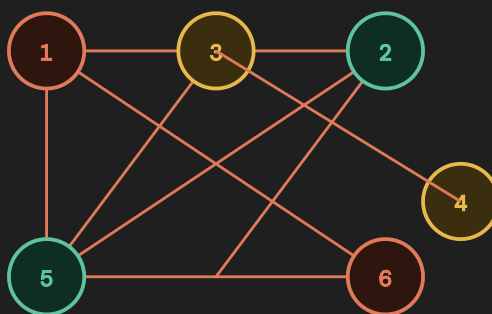
► MODEL ANSWER

1 Find all pairs within 150 miles (conflict = edge):

Pair	Distance	Conflict?
1–2	85	Yes (<150)

1-3	175	No
1-4	200	No
1-5	50	Yes
1-6	100	Yes
2-3	125	Yes
2-4	175	No
2-5	100	Yes
2-6	160	No
3-4	100	Yes
3-5	200	No
3-6	250	No
4-5	210	No
4-6	220	No
5-6	100	Yes

② **Edges (conflicts):** 1-2, 1-5, 1-6, 2-3, 2-5, 3-4, 5-6



Radio station conflict graph. Same colour = same channel assignment.

③ **Minimal colouring:** Station 1-2-5 form a triangle → need at least 3 colours. We can 3-colour:

Channel	Stations
Channel 1 (Terracotta)	1, 3

Channel 2 (Teal)	2, 6
Channel 3 (Amber)	4, 5

Verify: 1–2 ✓ different, 1–5 ✓ different, 1–6 ✓ different, 2–3 ✓ different, 2–5 ✓ different, 3–4 ✓ different, 5–6 ✓ different.

4 Adjacency list:

Station	Adjacent
1	2, 5, 6
2	1, 3, 5
3	2, 4
4	3
5	1, 2, 6
6	1, 5

♦ Chromatic number = 3 channels are required.

QUESTION 3

Consider the directed graph with nodes 1, 2, 3, 4. Is the relation reflexive, symmetric, transitive? List ordered pairs.

► MODEL ANSWER

1 **Read the graph:** From the figure, the directed edges (arrows) are:

- Self-loop at 1: (1, 1)
- Self-loop at 4: (4, 4)
- $1 \rightarrow 2$: (1, 2)
- $2 \rightarrow 1$: (2, 1)
- $3 \rightarrow 1$: (3, 1)
- $3 \rightarrow 4$: (3, 4)
- $4 \rightarrow 3$: (4, 3)

So the relation is $R = \{(1, 1), (4, 4), (1, 2), (2, 1), (3, 1), (3, 4), (4, 3)\}$.

- 2 (a) **Reflexive?** Every element must relate to itself: $(1, 1), (2, 2), (3, 3), (4, 4)$ must all be in R . We have $(1, 1)$ and $(4, 4)$ but NOT $(2, 2)$ or $(3, 3)$. → **NOT reflexive.**
- 3 (b) **Symmetric?** For every $(a, b) \in R$, we need $(b, a) \in R$. Check $(3, 1) \in R$: is $(1, 3) \in R$? No. → **NOT symmetric.**
- 4 (c) **Transitive?** For every $(a, b), (b, c) \in R$, we need $(a, c) \in R$. Check: $(3, 1), (1, 2) \in R$ — is $(3, 2) \in R$? No. → **NOT transitive.**

✦ **NOT reflexive, NOT symmetric, NOT transitive.**

$$R = \{(1, 1), (4, 4), (1, 2), (2, 1), (3, 1), (3, 4), (4, 3)\}$$

QUESTION 4

Relation $R = \{(a, b) \mid a \text{ divides } b\}$ on $A = \{1, 2, 3, 4, 6\}$. Reflexive, symmetric, transitive? Draw graph.

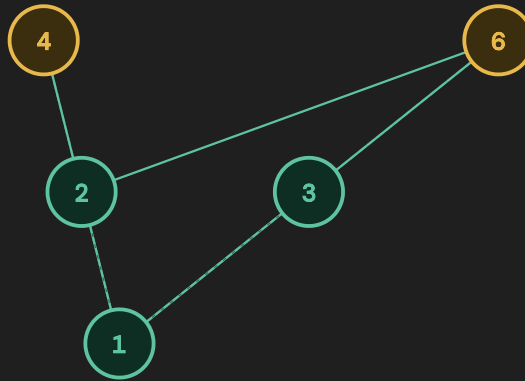
► MODEL ANSWER

- 1 List all ordered pairs (a, b) where $a \mid b$:

a	b values that a divides	Pairs
1	1,2,3,4,6	$(1,1), (1,2), (1,3), (1,4), (1,6)$
2	2,4,6	$(2,2), (2,4), (2,6)$
3	3,6	$(3,3), (3,6)$
4	4	$(4,4)$
6	6	$(6,6)$

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (6, 6)\}$$

- 2 (a) **Reflexive?** Every $a \in A$ divides itself ($a \mid a$). So $(1, 1), (2, 2), (3, 3), (4, 4), (6, 6) \in R$ ✓ → **YES, reflexive.**
- 3 (b) **Symmetric?** Does $a \mid b$ imply $b \mid a$? Counterexample: $1 \mid 2$ but $2 \nmid 1$. → **NOT symmetric.**
- 4 (c) **Transitive?** If $a \mid b$ and $b \mid c$, then $a \mid c$. This is a fundamental property of divisibility. → **YES, transitive.** (This relation is a partial order — reflexive, antisymmetric, transitive.)
- 5 (d) **Directed graph (Hasse-style diagram for divisibility):**



Dashed arrows = transitive closure (derived from composition)

Divisibility relation on $\{1, 2, 3, 4, 6\}$. Solid arrows show direct divisibility; dashed are implied by transitivity. Self-loops (reflexivity) omitted for clarity.

♦ Reflexive: YES | Symmetric: NO | Transitive: YES | (This is a partial order)

TUTORIAL 3

Primes and GCD

Q1(A) – PRIMALITY TESTS

Determine whether 667, 557, and 9269 are prime.

► MODEL ANSWER

Primality test: To test if n is prime, try dividing by all primes $p \leq \sqrt{n}$. If none divide n , then n is prime.

i $n = 667$: $\sqrt{667} \approx 25.8$. Primes to test: 2, 3, 5, 7, 11, 13, 17, 19, 23.

- $667 \div 2 = 333.5$ — No
- $667 \div 3$: digit sum = $6+6+7=19$ — No
- $667 \div 7 = 95.3$ — No
- $667 \div 11 = 60.6$ — No
- $667 \div 13 = 51.3$ — No
- $667 \div 17 = 39.2$ — No
- $667 \div 19 = 35.1$ — No

- $667 \div 23 = 29 \checkmark$ — **YES!** $23 \times 29 = 667$

→ **667 is NOT prime** ($= 23 \times 29$).

ii $n = 557$: $\sqrt{557} \approx 23.6$. Primes to test: 2, 3, 5, 7, 11, 13, 17, 19, 23.

- $557 \div 2$ — No (odd)
- $557 \div 3$: digit sum = $5+5+7=17$ — No
- $557 \div 5$ — No (doesn't end in 0 or 5)
- $557 \div 7 = 79.57$ — No
- $557 \div 11 = 50.6$ — No
- $557 \div 13 = 42.8$ — No
- $557 \div 17 = 32.8$ — No
- $557 \div 19 = 29.3$ — No
- $557 \div 23 = 24.2$ — No

→ **557 is PRIME.**

iii $n = 9269$: $\sqrt{9269} \approx 96.3$. Test primes up to 96: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89.

- $9269 \div 2$ — No (odd)
- $9269 \div 3$: digit sum = 26 — No
- $9269 \div 7 = 1324.1$ — No
- $9269 \div 11 = 842.6$ — No
- $9269 \div 13 = 713 \checkmark$ — **YES!** $13 \times 713 = 9269$, and $713 = 23 \times 31$

→ **9269 is NOT prime** ($= 13 \times 23 \times 31$).

♦ **667: Not prime (23×29) | 557: Prime | 9269: Not prime (13×23×31)**

Q1(B) — GCD AND LCM FROM PRIME FACTORIZATIONS

Find GCD and LCM: (i) $3^7 \cdot 5^3 \cdot 7^3$ and $2^{11} \cdot 3^5 \cdot 5^9$ (ii) $2^2 \cdot 3^3 \cdot 5^5$ and $2^5 \cdot 3^3 \cdot 5^2$ (iii) $2^2 \cdot 7$ and $5^3 \cdot 13$

► MODEL ANSWER

Rule: $\gcd = \prod p^{\min(\alpha, \beta)}$, $\text{lcm} = \prod p^{\max(\alpha, \beta)}$ over all primes.

i $a = 3^7 \cdot 5^3 \cdot 7^3$, $b = 2^{11} \cdot 3^5 \cdot 5^9$

Prime	Exp in a	Exp in b	min (GCD)	max (LCM)
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2	0	11	0	11
3	7	5	5	7
5	3	9	3	9
7	3	0	0	3

$$\text{gcd} = 3^5 \cdot 5^3 = 243 \times 125 = 30375$$

$$\text{lcm} = 2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3$$

(ii) $a = 2^2 \cdot 3^3 \cdot 5^5, b = 2^5 \cdot 3^3 \cdot 5^2$

$$\text{gcd} = 2^{\min(2,5)} \cdot 3^{\min(3,3)} \cdot 5^{\min(5,2)} = 2^2 \cdot 3^3 \cdot 5^2 = 4 \times 27 \times 25 = 2700$$

$$\text{lcm} = 2^5 \cdot 3^3 \cdot 5^5 = 32 \times 27 \times 3125 = 2,700,000$$

(iii) $a = 2^2 \cdot 7^1, b = 5^3 \cdot 13^1$. These share NO common prime factors.

$$\text{gcd} = 1 \quad (\text{they are relatively prime})$$

$$\text{lcm} = 2^2 \cdot 5^3 \cdot 7 \cdot 13 = 4 \times 125 \times 7 \times 13 = 45500$$

✦ (i) $3^5 \cdot 5^3$, $2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3$ | (ii) $\text{gcd}=2700, \text{lcm}=2,700,000$ | (iii) $\text{gcd}=1, \text{lcm}=45500$

Q1(C) – PAIRWISE RELATIVELY PRIME

Are (i) {21, 34, 55} and (ii) {14, 15, 21} pairwise relatively prime?

► MODEL ANSWER

Pairwise relatively prime means every pair from the set has $\text{gcd} = 1$.

(i) Factorize: $21 = 3 \times 7, 34 = 2 \times 17, 55 = 5 \times 11$.

- $\text{gcd}(21, 34) = 1 \checkmark$
- $\text{gcd}(21, 55) = 1 \checkmark$
- $\text{gcd}(34, 55) = 1 \checkmark$

→ YES, pairwise relatively prime.

(ii) Factorize: $14 = 2 \times 7, 15 = 3 \times 5, 21 = 3 \times 7$.

- $\gcd(14,15) = 1 \checkmark$
- $\gcd(14,21) = 7 \times$
- $\gcd(15,21) = 3 \times$

→ NO, not pairwise relatively prime.

♦ (i) YES | (ii) NO

Q1(D) – GCD(1000, 625) AND VERIFICATION

Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$. Verify that $\gcd \times \text{lcm} = 1000 \times 625$.

► MODEL ANSWER

① Prime factorise: $1000 = 2^3 \times 5^3$, $625 = 5^4$.

②
$$\gcd(1000, 625) = 2^{\min(3,0)} \times 5^{\min(3,4)} = 2^0 \times 5^3 = 125$$

③
$$\text{lcm}(1000, 625) = 2^{\max(3,0)} \times 5^{\max(3,4)} = 2^3 \times 5^4 = 8 \times 625 = 5000$$

④ Verification:

$$\gcd \times \text{lcm} = 125 \times 5000 = 625,000$$

$$1000 \times 625 = 625,000$$

✓

♦ $\gcd = 125$, $\text{lcm} = 5000$. Verified: $125 \times 5000 = 625,000 = 1000 \times 625 \checkmark$

Q2(A) – EUCLIDEAN ALGORITHM

Compute: (i) $\gcd(111, 201)$ (ii) $\gcd(252, 356)$ (iii) $\gcd(1000, 5040)$

► MODEL ANSWER

Euclidean Algorithm: $\gcd(a, b) = \gcd(b, a \bmod b)$, repeat until remainder = 0. The last non-zero remainder is the gcd.

(i) $\gcd(111, 201)$:

$$201 = 1 \times 111 + 90$$

$$111 = 1 \times 90 + 21$$

$$90 = 4 \times 21 + 6$$

$$21 = 3 \times 6 + 3$$

$$6 = 2 \times 3 + 0$$

Last non-zero remainder: $\gcd(111, 201) = 3$.

(ii) $\gcd(252, 356)$:

$$356 = 1 \times 252 + 104$$

$$252 = 2 \times 104 + 44$$

$$104 = 2 \times 44 + 16$$

$$44 = 2 \times 16 + 12$$

$$16 = 1 \times 12 + 4$$

$$12 = 3 \times 4 + 0$$

$\gcd(252, 356) = 4$.

(iii) $\gcd(1000, 5040)$:

$$5040 = 5 \times 1000 + 40$$

$$1000 = 25 \times 40 + 0$$

$\gcd(1000, 5040) = 40$.

♦ (i) $\gcd(111, 201) = 3$ | (ii) $\gcd(252, 356) = 4$ | (iii) $\gcd(1000, 5040) = 40$

Q2(B-D) – MIXED GCD PROBLEMS

Q2b: $p = 2^{53} \cdot 4^{11} \cdot 7^{13} \cdot 10^1$ and $q = 3^6 \cdot 11^5 \cdot 13^{12} \cdot 17^4$. Q2c: Product = $2^{73} \cdot 8^5 \cdot 27^{11}$, $\gcd = 2^3 \cdot 3^{45}$.

Q2d: Show $n^2 + 3n + 2$ is never prime for positive n .

► MODEL ANSWER

(b.i) Simplify p and q first: $p = 2^{53} \cdot (2^2)^{11} \cdot 7^{13} \cdot (2 \cdot 5)^1 = 2^{53+22+1} \cdot 5^1 \cdot 7^{13} = 2^{76} \cdot 5 \cdot 7^{13}$

$$q = 3^6 \cdot 11^5 \cdot 13^{12} \cdot 17^4.$$

p and q share NO common prime factors. $\gcd(p, q) = 1$.

b.i.i $\text{lcm}(p, q) = p \times q$ (since $\text{gcd} = 1$).

b.iii $\text{gcd}(p, q) \cdot \text{lcm}(p, q) = 1 \times pq = pq \checkmark$

c **lcm from product and gcd:** Simplify: $\text{product} = 2^{73} \cdot (2^3)^5 \cdot (3^3)^{11} = 2^{73+15} \cdot 3^{33} = 2^{88} \cdot 3^{33}$.
 $\text{gcd} = 2^3 \cdot 3^{45}$.

Using $pq = \text{gcd} \cdot \text{lcm}$:

$$\text{lcm} = \frac{pq}{\text{gcd}} = \frac{2^{88} \cdot 3^{33}}{2^3 \cdot 3^{45}} = 2^{85} \cdot 3^{-12}$$

$$\boxed{\text{lcm} = \frac{2^{85}}{3^{12}}}$$

d Show $n^2 + 3n + 2$ is never prime for positive integers n : Factor:

$$n^2 + 3n + 2 = (n + 1)(n + 2)$$

For any positive integer n , both $n + 1 \geq 2$ and $n + 2 \geq 3$. So $n^2 + 3n + 2$ is the product of two integers both ≥ 2 . A number is prime only if it has exactly one non-trivial factor — but here we have two, so **it is always composite**. The statement is **FALSE**.

✦ **Q2d:** $n^2 + 3n + 2 = (n + 1)(n + 2)$ is always composite for n . Statement is **FALSE**.
positive

TUTORIAL 12

Number Theory

QUESTION 1

Evaluate $155 \pmod{19}$ and $-221 \pmod{23}$.

MODEL ANSWER

$a \bmod m$ is the remainder r when a is divided by m , with $0 \leq r < m$.

1 $155 \bmod 19$: Divide: $155 \div 19 = 8$ remainder r . $8 \times 19 = 152$. So $155 - 152 = 3$.

$$155 = 8 \times 19 + 3 \implies 155 \equiv 3 \pmod{19}$$

- ② $-221 \bmod 23$: First: $221 \div 23 = 9$ remainder r . $9 \times 23 = 207$. So $221 = 9 \times 23 + 14$, meaning $221 \equiv 14 \pmod{23}$. For negative numbers: $-221 \equiv -14 \pmod{23}$. To make positive: $-14 + 23 = 9$.

$$-221 \equiv 9 \pmod{23}$$

Verify: $-221 = (-10) \times 23 + 9$. Indeed $-10 \times 23 = -230$ and $-230 + 9 = -221 \checkmark$.

$$\spadesuit \quad 155 \bmod 19 = 3 \quad | \quad -221 \bmod 23 = 9$$

QUESTION 2

Find $a \bmod m$ and $a \operatorname{div} m$ for: (i) $a = 9009, m = 223$; (ii) $a = -10101, m = 333$.

► MODEL ANSWER

Division algorithm: $a = q \cdot m + r$ where $q = a \operatorname{div} m$ and $r = a \bmod m$, with $0 \leq r < m$.

- ① $a = 9009, m = 223$. Given: $9009 \operatorname{div} 223 = 40$. So $q = 40$. Remainder:
 $r = 9009 - 40 \times 223 = 9009 - 8920 = 89$.

$$9009 \operatorname{div} 223 = 40, \quad 9009 \bmod 223 = 89$$

- ② $a = -10101, m = 333$. $-10101 \div 333 \approx -30.33$. Floor: $q = -31$ (floor of -30.33 is -31).
 $r = -10101 - (-31)(333) = -10101 + 10323 = 222$. Verify: $0 \leq 222 < 333 \checkmark$.

$$-10101 \operatorname{div} 333 = -31, \quad -10101 \bmod 333 = 222$$

$$\spadesuit \quad \text{(i) div}=40, \text{ mod}=89 \quad | \quad \text{(ii) div}=-31, \text{ mod}=222$$

QUESTION 3

With $a \equiv 4 \pmod{13}$ and $b \equiv 9 \pmod{13}$, find c with $0 \leq c \leq 12$ such that: (i) $c \equiv 9a \pmod{13}$, (ii) $c \equiv a + b \pmod{13}$.

► MODEL ANSWER

Key property: If $a \equiv a' \pmod{m}$, then $ka \equiv ka' \pmod{m}$ and $a + b \equiv a' + b' \pmod{m}$.

(i) $c \equiv 9a \pmod{13}$. Since $a \equiv 4$:

$$9a \equiv 9 \times 4 = 36 \pmod{13}$$

$$36 = 2 \times 13 + 10 \implies 36 \equiv 10 \pmod{13}$$

$$c \equiv 10$$

(ii) $c \equiv a + b \pmod{13}$. With $a \equiv 4$ and $b \equiv 9$:

$$a + b \equiv 4 + 9 = 13 \equiv 0 \pmod{13}$$

$$c \equiv 0$$

♦ (i) $c = 10$ | (ii) $c = 0$

QUESTIONS 4-8 – PROOFS

Prove or disprove Q4–Q7; Prove Q8.

▶ MODEL ANSWER

Q4 – Product of two even integers is a multiple of 4

(1) Let m and n be even integers. By definition, $m = 2j$ and $n = 2k$ for integers j, k .

(2)

$$mn = (2j)(2k) = 4jk$$

(3) Since jk is an integer, $mn = 4(jk)$ is divisible by 4.

♦ ✓ TRUE. Proof complete. The product of any two even integers is always a multiple of 4.

Q5 – If $a|b$ and $a|c$, then $a|(2b - 3c)$

(1) Let $a|b$ and $a|c$. Then $b = as$ and $c = at$ for integers s, t .

2

$$2b - 3c = 2(as) - 3(at) = a(2s - 3t)$$

3 Since $2s - 3t$ is an integer, $a \mid (2b - 3c)$. ✓

♦ ✓ TRUE. Proved by direct calculation.

Q6 – If $a \mid b$ and $b \mid a$, then $a = b$

1 **Disprove** with a counterexample: Let $a = 3$ and $b = -3$. Then $3 \mid -3$ ✓ and $-3 \mid 3$ ✓, but $3 \neq -3$.

2 The correct statement would be $|a| = |b|$ (i.e., $a = \pm b$).

♦ ✗ FALSE. Counterexample: $a = 3, b = -3$. We have $3 \mid -3$ and $-3 \mid 3$, but $3 \neq -3$.

Q7 – If $a \mid bc$, then $a \mid b$ or $a \mid c$

1 **Disprove** with a counterexample: Let $a = 4, b = 2, c = 6$. Then $bc = 12$ and $4 \mid 12$ ✓. But $4 \nmid 2$ and $4 \nmid 6$. So neither $a \mid b$ nor $a \mid c$ holds.

2 Note: The statement WOULD be true if a is prime (Euclid's lemma), but fails for composite a .

♦ ✗ FALSE. Counterexample: $4 \mid 12 = 2 \times 6$, but $4 \nmid 2$ and $4 \nmid 6$.

Q8 – Show $n^2 \equiv 0 \pmod{4}$ or $n^2 \equiv 1 \pmod{4}$ for all integers n

1 **Case 1: n is even.** Then $n = 2k$ for some integer k . So $n^2 = 4k^2 = 4(k^2)$. Thus $n^2 \equiv 0 \pmod{4}$.

2 **Case 2: n is odd.** Then $n = 2k + 1$ for some integer k . So:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1$$

Thus $n^2 \equiv 1 \pmod{4}$.

3 Every integer is either even or odd, and we have covered both cases. ✓

♦ ✓ PROVED. Every perfect square is either $\equiv 0 \pmod{4}$ (if n even) or $\equiv 1 \pmod{4}$ (if n odd).

